

WATROUS EAST, SK, Canada

WMO: 715110

Lat: 51.670N

Lon: 105.400W

Elev: 526

StdP: 95.17

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-30.0	-36.1	0.1	-32.8	-33.0	0.2	-29.9	13.1	-9.5	11.8	-9.0	3.4	300	0.731

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	29.7	18.5	27.7	18.0	26.0	17.1	21.2	26.6	19.8	24.9	18.7	23.6	5.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	15.1	23.9	18.1	13.9	22.3	16.8	12.8	20.9	64.1	26.5	59.3	24.8	55.2	23.6	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	10.2	9.1	-37.2	33.8	3.3	1.8	-39.6	35.1	-41.6	36.2	-43.4	37.2	-45.9	38.5		
			-37.3	23.8	3.3	1.3	-39.6	24.7	-41.6	25.4	-43.4	26.1	-45.7	27.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-15.1	-13.6	-6.8	3.2	10.4	15.4	17.9	16.9	11.9	3.9	-5.5	-12.7	
	DBStd	13.38	8.73	7.99	7.85	5.67	4.65	3.46	2.87	3.33	4.36	4.93	6.76	7.90	
	HDD10.0	3621	779	660	520	211	53	2	0	1	29	197	464	704	
	HDD18.3	5944	1038	893	778	456	249	99	43	68	195	449	714	963	
	CDD10.0	786	0	0	0	6	64	164	244	214	87	7	0	0	
	CDD18.3	68	0	0	0	0	2	12	29	23	3	0	0	0	
	CDH23.3	1129	0	0	0	3	74	176	367	382	126	3	0	0	
	CDH26.7	290	0	0	0	1	15	39	87	114	33	0	0	0	
Wind	WSAvg	4.4	4.6	4.5	4.7	4.9	5.1	4.5	3.8	3.8	4.3	4.5	4.5	4.3	
Precipitation	PrecAvg	414	17	10	15	27	47	84	66	54	39	26	14	16	
	PrecMax	723	36	20	38	62	133	172	144	189	127	74	47	35	
	PrecMin	225	3	2	2	0	9	16	19	9	3	1	2	4	
	PrecStd	103	9	5	9	19	26	37	32	39	27	18	10	9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.0	11.7	21.9	28.7	30.5	31.6	33.0	30.2	22.6	12.1	5.2	
		MCWB	2.3	2.0	6.6	11.4	14.7	17.6	21.4	18.9	16.8	13.7	7.3	2.4	
	2%	DB	2.5	1.7	7.3	18.9	25.4	27.5	28.9	29.8	27.1	18.0	7.7	2.8	
		MCWB	1.1	0.4	4.3	9.9	13.2	17.0	20.5	18.2	15.5	10.7	4.2	1.3	
	5%	DB	0.4	0.0	4.7	15.9	22.8	25.2	27.1	27.4	23.9	14.5	5.0	0.5	
		MCWB	-0.7	-0.8	2.7	8.8	12.2	16.0	19.3	17.8	14.4	8.8	2.8	-0.6	
	10%	DB	-2.7	-2.0	2.6	12.6	20.2	23.2	25.3	25.2	20.9	11.6	3.0	-1.9	
		MCWB	-3.3	-2.7	1.4	7.2	11.1	15.0	18.6	16.8	13.1	7.5	1.6	-2.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.3	6.9	12.7	16.5	21.4	24.1	21.9	18.7	13.9	7.6	2.7
			MCDB	3.9	3.5	10.6	19.9	22.9	27.0	28.7	28.6	26.3	21.4	11.8	4.6
		2%	WB	1.1	0.6	4.6	11.0	14.6	19.2	22.0	20.4	16.8	11.4	4.7	1.2
			MCDB	2.4	1.5	7.0	17.5	21.8	23.7	27.1	26.3	24.1	15.9	7.0	2.6
5%		WB	-0.8	-0.8	2.8	9.3	13.3	17.5	20.5	19.1	15.2	9.5	3.0	-0.5	
		MCDB	0.3	0.0	4.5	14.6	21.0	22.4	25.2	24.9	22.0	13.5	4.6	0.6	
10%		WB	-3.3	-2.7	1.5	7.5	12.1	16.2	19.3	17.9	13.8	8.0	1.6	-2.7	
		MCDB	-2.7	-2.1	2.6	12.0	18.6	21.1	23.9	23.1	19.9	11.5	3.0	-1.9	
Mean Daily Temperature Range	5% DB	MDBR	9.8	9.9	9.7	11.7	13.8	11.8	12.4	13.7	13.7	11.2	9.2	9.2	
		MCDBR	11.1	10.4	10.5	18.0	18.1	15.8	15.9	18.3	19.8	17.1	12.1	11.0	
		MCWBR	9.9	9.3	7.9	10.0	8.2	7.3	8.5	8.2	9.4	10.1	8.9	9.6	
		MCDDBR	10.9	10.1	10.2	16.6	15.6	13.2	14.2	15.0	17.3	15.5	11.5	11.0	
	5% WB	MCWBR	9.9	9.2	7.9	9.9	7.8	7.0	8.5	8.0	8.8	9.6	8.7	9.7	
Clear-Sky Solar Irradiance	taub	0.250	0.263	0.293	0.328	0.342	0.355	0.361	0.365	0.316	0.284	0.251	0.233		
	taud	2.304	2.276	2.349	2.388	2.397	2.408	2.390	2.373	2.495	2.545	2.480	2.358		
	Ebn at Noon	780	871	910	912	911	899	886	862	877	848	792	752		
	Edh at Noon	62	85	97	107	111	111	111	107	85	66	52	49		
All-Sky Solar Radiation	RadAvg	1.19	2.24	3.73	4.82	5.90	6.00	6.28	5.08	3.74	2.17	1.09	0.84		
	RadStd	0.11	0.15	0.32	0.40	0.49	0.42	0.36	0.36	0.31	0.30	0.15	0.09		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page